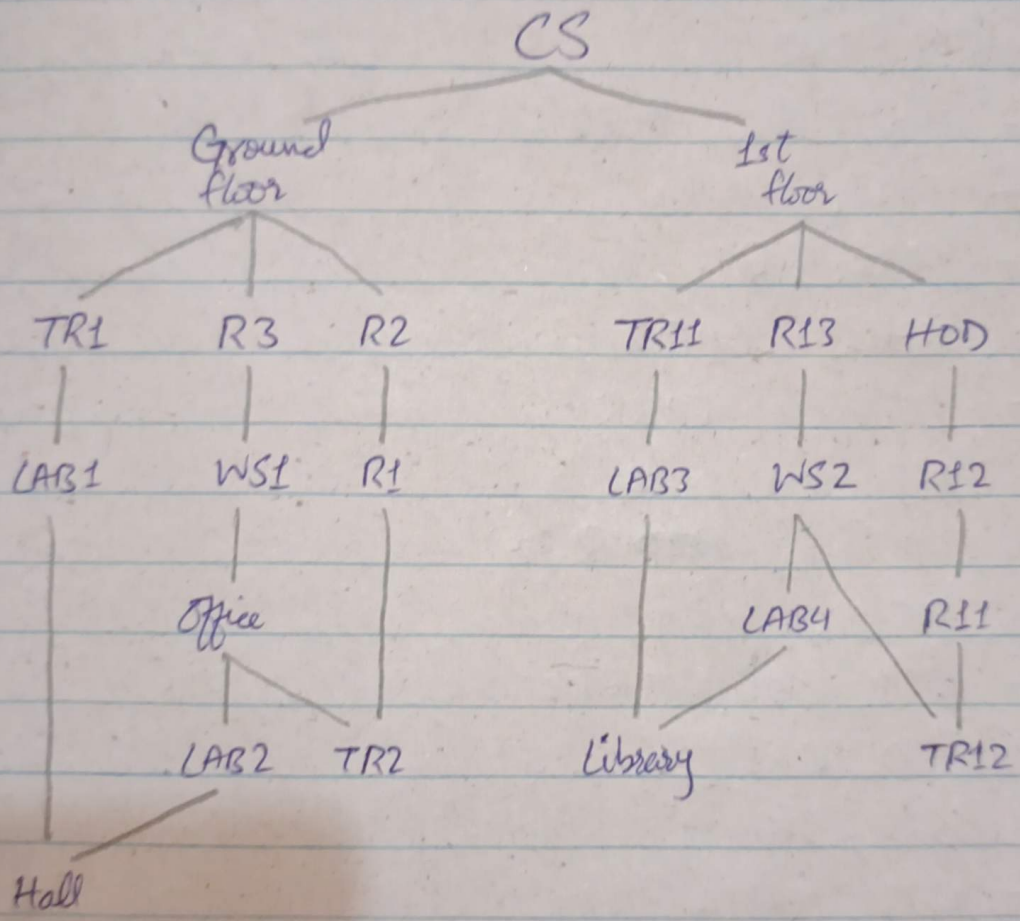
LAB Activity 5

Graph of CS Department :-



Transition Table :-

CS — Ground floor, 1st floor
Ground floor — CS, TR1, R3, R2
1st floor — CS, TR11, R13, HOD
TR1 — Ground floor, LAB 1
R3 — Ground floor, WS1

R2 — Ground floor, R1
LAB1 — TR1, Hall
WS1 — R3, Office
R1 — R2, TR2
Office — WS1, LAB2, TR2
LAB2 — Office, Hall
TR2 — Office, R1
Hall — LAB1, LAB2
TR11 — 1st floor, LAB3
R13 — 1st floor, WS2
HOD — 1st floor, R12
LAB3 — TR11, Library
WS2 — R13, LAB4, TR12
R12 — HOD, R11
LAB4 — WS2, Library
R11 — R12, TR12
Library — LAB3, LAB4

Problem Formulation :-

Initial State => LAB3
Goal State => LAB 1
State Space => All possible routes & direction in CS Department.

~~Transition~~

Action => Move from one possible state to an another state

Transition Model => Purpose to move state

Path Cost => Distance

Path LAB3 to LAB1 (BFS Table):-

Initialization:-

Q = [LAB3]

V = [LAB3]

Deq LAB3 & expand

Q = [TR11]

V = [LAB3, TR11]

Deq TR11 & expand

Q = [1st floor]

V = [LAB3, TR11, 1st floor]

Deq 1st floor & expand

Q = [CS]

V = [LAB3, TR11, 1st floor, CS]

Deq CS & expand

CamScanner

Q = [Ground floor]

V = [LAB3, TR11, 1st floor, CS, Ground floor]

Deq Ground floor & expand

Q = [TR1]

V = [LAB3, TR11, 1st floor, CS, Ground floor, TR1]

Deq TR1 & expand

Q = [LAB1]

V = [LAB3, TR11, 1st floor, CS, Ground floor, TR1, LAB1]

Shortest Path

LAB 3

↓

TR11

↓

1st Floor

↓

CS

↓

Ground Floor

↓

TR1

↓

LAB1